Shawn Jiang

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EDUCATION

University of Toronto Graduation: May 2028

*Honours Bachelor of Science in **Computer Science and Statistics***

- **Relevant Coursework:** Operating Systems, Databases, Machine Learning, Probability Theory, Complexity Theory

EXPERIENCE

Huawei Canada Aug. 2026 – Present

Software Engineer Intern (Rust, Python, AI Agents, Databases)

- Working on data agents in the **Distributed Data Storage and Management Lab**, supervised by Tianqing Wang.
- Researching grounding methods, and architectures for more efficient data agents. Aiming for conference publication.
- Built an offline annotation pipeline for clustering and deriving entity relations, resulted in **+7%** increase on **KramaBench**.
- Building agent harness and control mechanisms, for **observability**, and autonomous analysis in heterogeneous data-lakes.

Lee Language Lab Apr. 2026 – Present

Undergraduate Researcher (Python, PyTorch, RL, LLMs)

- Researching enhanced methods for pluralistic reasoning in LLMs, supervised by Dr. Annie Lee and Dr. Enas Altarawneh.
- Fine-tuned models using LoRA, and RL methods to enable structural interpretation and reasoning on graph structures.
- Designed a test-time dissimilarity injection mechanism, increasing OvertonBench evaluation metric from **0.48 to 0.54**.

Fanhan Investment Management Co., Ltd. May 2024 – Aug. 2024

Data Science Intern (Python, Pandas, Scikit-Learn)

- Constructed factor-adjusted portfolios using Fama-French and CAPM regression models in Scikit-Learn, identifying overexposed risk factors and rebalancing allocations to reduce tracking error by **18%**.

University of Toronto May 2025 – Sep. 2025

Software Engineer (Python, React, FastAPI, PostgreSQL, GCP)

- Architected and shipped a full-stack peer assessment platform in React, FastAPI, and PostgreSQL, enabling instructors to benchmark student-group performance. Currently in use by **200+ students**.
- Engineered a drag-and-drop ingestion pipeline with CSV parsing and schema validation, reducing manual data entry time by **80%**; deployed a serverless REST API on GCP Cloud Run behind a reverse proxy. Supervised by **Prof. Sohee Kang**.

Remeda Apr. 2026 – Jul. 2026

Software Engineer (Python, React, Deep Learning)

- Worked on job scheduling, CI/CD **build deployment**, built features integrating auscultation audio and medical imaging with clinical text for diagnostic reasoning models.

Seequent Jan. 2025 – Apr. 2025

Software Engineer Intern (C#, .NET)

- Developed and shipped production backend features in C#/.NET for geophysical modeling software used by **10,000+** geoscientists worldwide, including object conversion modules on the Evo cloud platform.
- Triaged and resolved **50+ bugs** across automated test suites in an **agile** sprint cadence, collaborating with QA engineers and senior developers to ship fixes to production within the same sprint.

PROJECTS

InterSync | *TypeScript, Node.js, PostgreSQL, WebRTC, CI/CD*

- Built a real-time collaborative interview platform with **WebRTC** video conferencing and live code execution; **automated testing** and deployment via a **CI/CD** pipeline built with GitHub Actions and **Docker**.
- Built resume chunking and **semantic search** using Gemini text embeddings and HNSW graphs for similarity search.

Chord Transformer | *Python, PyTorch, Transformer, NLP*

- Built a melody conditioned decoder-only transformer for chord harmonization, achieved chord-root perplexity 2.11.
- Engineered a decomposed tokenization scheme, reducing vocabulary size while preserving harmonic structure.
- Implemented **multi-head causal self-attention** with fused QKV projections and an autoregressive decoding pipeline.

TECHNICAL SKILLS

Languages: Python, SQL, Haskell, C, C++ TypeScript, Bash, R, Java

Libraries/Frameworks: FastAPI, React, Node.js, PyTorch, Scikit-Learn, LangGraph, Pandas, NumPy, Matplotlib

Tools: Linux, Git, Docker, GCP, PostgreSQL, MongoDB, Firebase, SQLite, Postman, Jupyter Notebook, SLURM